From Human Vision to Machine Vision

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1 Summary

Workshop title	From Human Vision to Machine Vision
Acronym	HVMV
Edition (1st, 2nd, ...)	1st
Keywords	Visual Computing, Human Vision, Machine/Computer Vision
Primary contact name and email	Kenny Chen (kennychen@nyu.edu)
Half day, full day, 90-minute, or 3/4 day	Half day
Scheduling preference (Sun or Mon-Thur)	Mon-Thur
Sketch submissions?	No
Special requests	None

2 Objective and Background

Visual computing encompasses the full range of technologies and methodologies involved in rendering, displaying, reconstructing, modeling, and recording visual data. Among the many subfields within visual computing, there has been a growing interest in two subfields in particular: human vision and machine vision. Human vision research (covering topics such as augmented and virtual reality, real-time high-fidelity rendering, interface design, and human-computer interaction) focuses on understanding how human vision and cognition functions and using that knowledge to improve

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the user experience through realism, immersion, and responsiveness. In contrast, machine vision research (spanning topics such as 3D reconstruction, scene understanding, neural rendering, visual-language models, SLAM, and action recognition) focuses on building computational systems that complete a visual perception task, and may aim to mimic the structure or functionality of biological vision systems.

Despite their shared roots in visual perception, these two research communities tend to be separate, often employing different methodologies and adopting different philosophies that guide their research. Both, however, study facets of the same fundamental phenomenon: the visual system. Our workshop posits that because both fields are effectively studying the same underlying concept, there is significant untapped potential in their convergence. Indeed, our experience as researchers working at the intersection of these two subfields shows that work in machine vision benefits greatly from insights in human vision research, and vice versa. Establishing stronger connections between machine and human vision researchers will lead to more holistic approaches, where advances in one field inform and inspire the other. For example, machine vision models may reveal computational insights into human perception, while studies of visual cognition can suggest new directions for more efficient or interpretable machine vision architectures. Beyond just the methodologies used in either field, researchers from these two communities have differing philosophies regarding how problems within visual computing should be solved. Should a machine vision system be built from the ground up, based on studies of human vision? Or should black-box (e.g. deep learning) models be loosely inspired by these human vision systems? Answering questions like these is currently difficult due to the minimal cross-talk between the two fields.

This workshop aims to facilitate dialogue and collaboration between researchers and practitioners in machine and human vision. By exploring common challenges, methodologies, and philosophical approaches, ranging from biologically grounded models to data-driven deep learning frameworks, we hope to spark interdisciplinary discussion that drives innovation across the broader field of visual computing.

3 Workshop Topics

We will coordinate with the invited speakers to center their keynote talks and the discussions around the following topics:

- **Neural Networks as Models of Human Vision:** Modern machine vision methodologies like deep neural networks are increasingly utilized by neuroscientists and computer scientists alike to construct models of human visual processing [1, 5, 9, 16, 17]. Studying how and when machine systems match human visual perception performance can reveal new computational models for how the brain might be implementing various visual functions, deepening our understanding of human visual cognition.
- **Perceptually-Optimized Rendering and Displays:** Rendering pipelines and display technologies for immersive displays are increasingly designed around the user's subjective perceptual experience. For example, researchers have developed rendering methods to optimize display power [2] and rendering efficiency [14] without detracting from the user's perceptual fidelity, to optimize rendering to produce stimuli that drive the human visual system in more natural ways [4], and to optimize distortion correction in immersive displays [8]. This focus enables more efficient pipelines and more faithful perceptual experiences in research and applications.
- **Improving Interpretability of Machine Vision Systems:** Empirical research has shown that by directly comparing machine vision systems to human vision, we can gain insights into the capabilities and limitations of neural networks for visual perception tasks and develop more robust machine vision systems [7, 10]. Furthermore,

human vision-inspired metrics [12, 13, 18] and architectures [3, 11] have led to significant advancements in visual computing.

- **Computer Graphics for Studying Human Vision:** Understanding the functionality of human vision requires exposing observers to specific, carefully-designed visual stimuli that test particular aspects of human vision. Typically, vision scientists use very simple stimuli, sacrificing realism for experimental control. With computer graphics, we can render more complex, naturalistic stimuli without sacrificing experimental control, which enables researchers to study more complex, high-level aspects of visual perception [6, 15].
- **Further Opportunities for Interdisciplinary Collaboration:** We will discuss new opportunities and avenues for collaboration between human and machine vision researchers. Are there existing long-standing problems that researchers believe can be solved with the help of researchers from another field? What challenges do you anticipate from collaborating with researchers on the other side of visual computing (e.g., communication issues, methodological traditions)?

4 Format and logistics

The workshop will consist of two main phases: keynote talks and a moderated discussion between the invited speakers and the audience.

Keynote Talks: The first portion of the workshop will feature 30-minute keynote talks by each our invited speakers, as well as two 20-minute Q&A sessions. The speakers will be invited to present their perspectives on the relationship between human and machine vision in visual computing, and to discuss how these two domains might mutually advance through the exchange of ideas and methodologies. Q&A sessions will take place after the second and the fourth keynote talks.

Moderated Discussion: The second portion of the workshop is a moderated, interactive discussion between the audience and a panel of the invited speakers. The main goal of our workshop is to foster collaboration and cross-pollination of ideas between human and machine vision researchers within visual computing. Therefore, the open discussion is crucial for allowing members of the community to share their ideas and ask thought-provoking questions that may inspire researchers to view their research questions under a new light. We may also solicit questions and comments from the audience via Slido[1] and via a microphone in the room. To minimize downtime where nothing is being discussed (e.g., if the audience does not have any questions), we will prepare a set of thought-provoking questions beforehand. Our role as moderators will be to ensure that the audience has ample opportunity to participate in the discussion and that every panelist has an equal amount of time to share their thoughts.

4.1 Schedule

The workshop will run for half day (3.25 hours). The first 2.75 hours will include talks and Q&A discussions. The last 30 minutes will be left for open-ended and moderated discussion between the invited speakers and the audience/moderators. The detailed schedule is as follows:

[1] https://www.slido.com/

Table 1. Event Schedule

Time Slot	Duration	Activity
9:00am – 9:05am	5 min	Opening remarks
9:05am – 9:35am	30 min	Talk 1
9:35am – 10:05am	30 min	Talk 2
10:05am – 10:20am	15 min	Q&A discussion
10:20am – 10:50am	30 min	Talk 3
10:50am – 11:20am	30 min	Talk 4
11:20am – 11:40am	20 min	Q&A discussion
11:40am – 11:45am	5 min	Break / Setup for open discussion
11:45am – 12:15pm	30 min	Moderated discussion

4.2 Sketch submission

Our workshop will not include paper submissions.

4.3 Special requests

We do not make any special requests.

5 Organizers

- **Kenneth Chen** (PhD Student, New York University)
  Kenneth Chen is a PhD student at New York University, advised by Prof. Qi Sun. He earned an MS in computer science from the same institution, and a BA in compuer science from UC Berkeley. His work focuses on visual perception as it applies to graphics and computational displays. He has worked at Meta Reality Labs and the University of Cambridge as an intern.
- **Dr. Niall L. Williams** (Postdoctoral Researcher, University of Zaragoza)
  Niall Williams is a postdoctoral researcher at the University of Zaragoza and an incoming assistant professor at Barnard College. He has interned at Meta Reality Labs and Nvidia, where he worked on perceptual constraints for immersive varifocal displays, perception of digital humans, and video artifacts. He earned a PhD and MS in Computer Science from the University of Maryland, and a BS in Computer Science from Davidson College. His work studies computational methods to improve the user's perceptual experience in immersive displays and how to use computer graphics to probe the capabilities and limitations of human vision.
- **Dr. Colin Groth** (Postdoctoral Researcher, New York University)
  Colin Groth is a postdoctoral researcher at New York University in the Immersive Computing Lab. He received his MSc and Ph.D. in Computer Science from Technische Universität Braunschweig, Germany, in 2020 and 2024, respectively. He was also working as a Postdoc in the Computer Graphics group of Prof. Hans-Peter Seidel at Max Planck Institute for Informatics. His research interests include perceptual-driven techniques, particularly for mitigating cybersickness in VR. With his methods and research, he also likes to encourage cross-disciplinary collaboration and develop applications for real-time rendering, image processing, and augmented reality.

- **Jenna Kang** (PhD Student, New York University)
  Jenna Kang is a PhD student at New York University, advised by Prof. Qi Sun. She received her B.S. in computer science from the Georgia Institute of Technology. Her research interests combine visual perception, psychophysics, and computer graphics. She is particularly interested in applying vision science to computer graphics, with a focus on perceptually grounded evaluation of visual quality in generative AI and video synthesis. Broadly, her work studies visibility in graphics across multiple perceptual levels, from detectability and artifact salience to attention, gaze, and visual memorability, with the goal of leveraging visual perception to inform graphics system design and understand how perception influences human behavior and task performance.
- **Dr. Alexandre Chapiro** (Imaging Architect and Senior Staff Research Scientist, Meta Reality Labs)
  Alexandre is an imaging architect and senior staff research scientist at Meta's Imaging Experiences Architecture team in Reality Labs. Previously, he worked in the Applied Perception Science team at Meta, Core Display Incubation team at Apple, the Applied Vision Science team at Dolby Laboratories, and the Stereo and Displays group at Disney Research Zurich. He earned a PhD from the Computer Graphics Laboratory at ETH Zurich, and holds MS and BS degrees in Mathematics. Alexandre is interested in solving novel problems for industry applications, touching on perception, computer graphics, computational display, and psychophysics. Prior work involved perceptual difference metrics, brightness and color, stereo 3D, the perception of faces, and display topics like virtual and augmented reality, frame rate, high dynamic range and more.
- **Prof. Qi Sun** (Associate Professor, New York University)
  Qi Sun is an associate professor at New York University. Before joining NYU, he was a research scientist at Adobe Research. He received his PhD at Stony Brook University. His research interests lie in VR/AR, perceptual computer graphics, computational displays, and applied perception. He is a recipient of the IEEE Virtual Reality Best Dissertation Award. With colleagues, his research has been recognized as several best paper and honorable mention awards at ACM SIGGRAPH, IEEE ISMAR, IEEE VR, IEEE VIS, and ACM SAP.

5.1 Organizers' experience and background

The organizers of this workshop are researchers whose interests lie at the intersection of computer graphics, applied (human) vision science, and machine learning. We have extensive experience with publishing in and reviewing for SIGGRAPH and in visual computing more broadly, and we have a presence within the vision science community as well. Additionally, Dr. Williams and Dr. Groth have experience running successful workshops in other top conference venues (IEEE International Symposium on Mixed and Augmented Reality and IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)).

6 Invited Speakers

Prof. Rafał Mantiuk | `https://www.cl.cam.ac.uk/~rkm38/index.html` | Confirmed
Relevance to workshop: Prof. Mantiuk is a professor at the University of Cambridge, where he works on topics spanning psychophysics, applied visual perception, computational displays and graphics, and more. He continues to publish pioneering work on high dynamic range imaging, image and video quality assessment metrics, computational displays, and perceptual rendering techniques. Furthermore, his recent research includes work that directly evaluates whether computer vision foundation models (such as DINO and OpenCLIP) align with biological vision. His expertise will be invaluable for attendees seeking to understand both the convergence and the persistent gaps between artificial and

biological visual systems.

Prof. Laurie M. Wilcox | https://www.wilcoxlab.yorku.ca/current-lab-24 | Confirmed
Relevance to workshop: Prof. Wilcox is the York Research Chair in 3D Vision at York University, where she specializes in the neural mechanisms of depth perception in both real and virtual environments. Her work often utilizes virtual and augmented reality devices to probe the mechanics of human vision, directly aligning with one of the workshop's goal of understanding how human vision functions to improve user experience for computer graphics. Her research serves as a prime example of the benefits of interdisciplinary collaboration, demonstrating how insights from biological vision can guide the development of visual computing systems. Furthermore, having served as the president of the Vision Sciences Society (the world's premier organization for vision research and its applications) and having collaborated directly with industry leaders (e.g. Meta, Qualcomm), Prof. Wilcox is an ideal authority to bridge the gap between rigorous perceptual science and practical display engineering.

Dr. Aaron Hertzmann | https://aaronhertzmann.com | Confirmed
Relevance to workshop: Dr. Hertzmann is a Principal Scientist at Adobe Research and an Affiliate Professor at the University of Washington. His research interests are in a unique space at the convergence of machine learning, the psychology of art, and computational photography. He is a pioneer in key graphics techniques like non-photorealistic rendering and computational imaging, and has developed algorithms that bridge the gap between physical camera optics and human aesthetic perception. This directly supports our workshop's goal of fostering crosstalk between machine vision and human vision studies. His theory for perspective perception in images and his work on image stylization demonstrate how computational systems can be designed to mimic biological seeing and artistic abstraction rather than strictly adhering to photorealistic physics. His perspective will be invaluable for discussing how visual computing systems can be optimized for human observers.

Dr. Richard Zhang | http://richzhang.github.io | Confirmed
Relevance to workshop: Dr. Zhang is a Senior Research Scientist at Adobe Research, where his work spans computer vision, deep learning, and graphics. He is the recipient of the 2017 Adobe Research Fellowship and was recognized as an Innovator Under 35 by MIT Technology Review in 2023. He is a leader in the evaluation of deep learning models through the lens of human perception, including developing the Learned Perceptual Image Patch Similarity (LPIPS) metric and the DreamSim metric. His research demonstrated that the internal representations of deep neural networks—even those trained on standard classification tasks—align surprisingly well with human perceptual judgments. By bridging the gap between pixel-level differences and subjective human experience, his work provides a critical foundation for discussing how machine vision architectures can be optimized to better reflect biological vision.

6.1 Diversity

a) Diversity of the organizing committee is being addressed by having representation across a broad range of career stages as well as representation from both academia and industry. Our organizing team consists of an early-stage PhD student, a late-stage PhD student, two postdoctoral researchers, a mid-stage professor, and a senior researcher from industry.

b) Diversity of the speakers was addressed by recruiting participants across a range of scientific disciplines and from both academia and industry. We made a strong effort to achieve gender balance among the invited speakers; however, some of the women we invited were unable to participate due to scheduling constraints. The final speaker lineup therefore reflects both disciplinary breadth and gender representation.

7 Broader impacts

Beyond enhancing the state-of-the-art in applied perception and machine vision for computer graphics, the convergence of these fields offers significant benefits to basic visual science research, perceptual psychology, and accessible technology.

- **Complex, Natural Stimuli:** By improving visual computing capabilities (via high-fidelity, perceptually-optimized rendering and display systems), researchers can develop highly controlled experiments using complex, real-world stimuli that are more naturalistic than the simple, traditional stimuli (such as Gabor patches) conventionally used in vision science. This opens the door for a deeper and more ecologically valid investigation into the underlying mechanisms of human vision.
- **Computational Models of Perception:** The methodologies and models developed within machine vision, including advanced neural networks (like those used in generative modeling and image synthesis), are increasingly being adapted by vision scientists to create sophisticated, functional models of human visual processing. The crosstalk fostered by this workshop will accelerate the development and validation of complex, biologically plausible computational models, providing new theoretical frameworks for understanding visual perception.
- **Accessibility and Assistive Technology:** Closing the gap between human and machine perception will make it easier for researchers and engineers to build assistive technologies that are more compatible and aligned with human perception. A better understanding of the capabilities and limitations of human vision enables us to create machine vision systems that are better able to complement humans and "fill the gaps" in human vision. For example, bionic vision to restore sight to blind individuals in a manner that is congruent with how the brain processes visual input.

7.1 Social considerations

- **Increased Understanding of Models:** AI is becoming more and more ingrained into all aspects of society, its influence on society increases. Consequently, it is more common day by day for laypeople to interact with AI models in their everyday lives. One of the goals of this workshop is to stimulate research that improves our understanding of complex, large foundation models that are becoming more prominent. An improved understanding of large computer vision systems will make it easier for scientists to communicate with laypeople about how these systems work, removing the "black box" stigma that pervades the current impression of AI. Thus, one potential consideration for this workshop is how our understanding of large machine vision systems can influence the cultural attitude towards AI in society. Furthermore, improved understanding of machine vision systems can help to mitigate algorithmic biases that arise from training data and architectural inductive biases.

7.2 Ethical considerations

- **Interpretability and Trust (The "Black Box" Problem):** The opacity of modern machine vision systems creates a significant trust gap, particularly when these technologies are deployed in safety-critical applications. By comparing these "black box" models to the relatively well-mapped functions of the human visual system, this workshop aims to promote the development of interpretable architectures that can be trusted in high-stakes environments.
- **Manipulation and Deepfakes:** As we develop graphics technologies that are perceptually optimized to match human visual limitations, we inadvertently lower the barrier for creating imagery that is indistinguishable from reality (e.g. deepfakes). We must address the ethical implications of these capabilities, specifically regarding their potential abuse in generating hyper-realistic deepfakes and spreading visual misinformation that exploits the vulnerabilities and limitations of the human visual system.
- **Algorithmic Bias vs. Human Bias:** While human vision is subject to innate cognitive biases, machine vision inherits statistical biases from its training data. A key ethical consideration for this workshop is to distinguish between biologically innate visual priors (which are universal) and learned data biases (which are often discriminatory). We must be careful not to confuse the two when using neural networks as models of human vision, as doing so risks naturalizing societal prejudices as "biological facts."

8 Relationship to previous workshops

The most similar workshop to ours is a workshop on applied human vision for augmented and virtual reality (https://www.pergravar-work.shop/2025/) at the 2026 IEEE Conference on Virtual Reality and 3D User Interfaces, run by Dr. Groth (one of the organizers for our proposed SIGGRAPH workshop). However, this workshop focused solely on work related to human visual perception for VR/AR, while our proposed workshop aims to bridge human and and machine vision. We are not aware of any other related workshops that have recently been held at computer science conferences.

References

[1] Yancheng Cai, Fei Yin, Dounia Hammou, and Rafal Mantiuk. Do computer vision foundation models learn the low-level characteristics of the human visual system? In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 20039–20048, 2025.

[2] Kenneth Chen, Thomas Wan, Nathan Matsuda, Ajit Ninan, Alexandre Chapiro, and Qi Sun. Pea-pods: Perceptual evaluation of algorithms for power optimization in xr displays. *ACM Transactions on Graphics (TOG)*, 43(4):1–17, 2024.

[3] Minkyu Choi, Kuan Han, Xiaokai Wang, Yizhen Zhang, and Zhongming Liu. A dual-stream neural network explains the functional segregation of dorsal and ventral visual pathways in human brains. *Advances in Neural Information Processing Systems*, 36:50408–50428, 2023.

[4] Steven A Cholewiak, Gordon D Love, Pratul P Srinivasan, Ren Ng, and Martin S Banks. Chromablur: Rendering chromatic eye aberration improves accommodation and realism. *ACM Transactions on Graphics (TOG)*, 36(6):1–12, 2017.

[5] Colin Conwell, Jacob S Prince, Kendrick N Kay, George A Alvarez, and Talia Konkle. A large-scale examination of inductive biases shaping high-level visual representation in brains and machines. *Nature communications*, 15(1):9383, 2024.

[6] Roland W Fleming, Antonio Torralba, and Edward H Adelson. Specular reflections and the perception of shape. *Journal of vision*, 4(9):10–10, 2004.

[7] Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A Wichmann, and Wieland Brendel. Imagenet-trained cnns are biased towards texture; increasing shape bias improves accuracy and robustness. In *International conference on learning representations*, 2018.

[8] Phillip Guan, Olivier Mercier, Michael Shvartsman, and Douglas Lanman. Perceptual requirements for eye-tracked distortion correction in vr. In *ACM SIGGRAPH 2022 Conference Proceedings*, pages 1–8, 2022.

[9] Dounia Hammou, Yancheng Cai, Pavan Madhusudanarao, Christos G Bampis, and Rafał K Mantiuk. Do image and video quality metrics model low-level human vision? *arXiv preprint arXiv:2503.16264*, 2025.

[10] Anne Harrington, Vasha DuTell, Ayush Tewari, Mark Hamilton, Simon Stent, Ruth Rosenholtz, and William T Freeman. Exploring perceptual straightness in learned visual representations. In *The Eleventh International Conference on Learning Representations*, 2022.

[11] Yann LeCun and Yoshua Bengio. Convolutional networks for images, speech, and time series. *The handbook of brain theory and neural networks*, 1998.

[12] Rafał K Mantiuk, Gyorgy Denes, Alexandre Chapiro, Anton Kaplanyan, Gizem Rufo, Romain Bachy, Trisha Lian, and Anjul Patney. Fovvideovdp: A visible difference predictor for wide field-of-view video. *ACM Transactions on Graphics (TOG)*, 40(4):1–19, 2021.

[13] Rafal K Mantiuk, Param Hanji, Maliha Ashraf, Yuta Asano, and Alexandre Chapiro. Colorvideovdp: A visual difference predictor for image, video and display distortions. *arXiv preprint arXiv:2401.11485*, 2024.

[14] Anjul Patney, Marco Salvi, Joohwan Kim, Anton Kaplanyan, Chris Wyman, Nir Benty, David Luebke, and Aaron Lefohn. Towards foveated rendering for gaze-tracked virtual reality. *ACM Transactions On Graphics (TOG)*, 35(6):1–12, 2016.

[15] Mohammad R Saeedpour-Parizi, Niall L Williams, Tim Wong, Phillip Guan, Dinesh Manocha, and Ian M Erkelens. Perceptual Thresholds for Radial Optic Flow Distortion in Near-Eye Stereoscopic Displays. *IEEE Transactions on Visualization and Computer Graphics*, 30(5):2570–2579, 2024.

[16] Felix A Wichmann and Robert Geirhos. Are deep neural networks adequate behavioral models of human visual perception? *Annual review of vision science*, 9(1):501–524, 2023.

[17] Daniel LK Yamins, Ha Hong, Charles F Cadieu, Ethan A Solomon, Darren Seibert, and James J DiCarlo. Performance-optimized hierarchical models predict neural responses in higher visual cortex. *Proceedings of the national academy of sciences*, 111(23):8619–8624, 2014.

[18] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018.